Demand Forecasting & Inventory Optimisation

Fresh Produce Retail Dataset

Operations & Decisions · Group Project · 2025

1. Data Preparation & Aggregation

The raw dataset contained 72,409 entries across 6 columns, covering a continuous window of 729 days per product (verified for Cucumber). Individual rows mapped to SKUs that shared a common parent product code (ARTNRVIER), meaning multiple size or packaging variants appeared on the same date — which would severely undercount true daily demand if naively deduplicated.

Aggregation logic

- Daily demand (C520_SOLDPIECES) was summed across all SKU variants of the same parent product and date.
- Retail price (C520_RETAILPRICE) was averaged across variants to obtain a representative daily price.
- Discount flag (DISCOUNT) was taken as the daily maximum — any variant being promoted flags the whole product as discounted.

Product selection – Top 10 by total volume

Loose Red Pepper, Aubergine, Cucumber, Spring Onions, Iceberg Lettuce, Broccoli, Wild Rocket, Fresh Spinach, Mixed Peppers, Family Pack Tomatoes.

Time-series construction

- Each product was reindexed to a fully continuous daily calendar (no gaps over 729 days).
- Missing demand days → filled with 0 (true zero-sales or store closed).
- Missing prices → forward-filled (last known price carried forward).
- Missing discount flags → set to 0 (no promotion assumed).

Limitation: Zero-filling assumes no unrecorded sales. Forward-filling prices assumes no missed promotions — reasonable but imperfect. The 4 residual NaNs in ARTNRVIER for Cucumber are non-critical (identifier column only, not used in modelling).

2. Demand Forecasting Results

Two models were evaluated per product using a strict time-based 80/20 train-test split (no data leakage):

- Naive Baseline – tomorrow's forecast = today's actual demand (lag_1).
- Random Forest Regressor (100 trees, max depth 10) – features: lag_1, lag_7, rolling_mean_7, day-of-week, month, retail price.

Table 1 – Mean Absolute Error (MAE) by product and forecasting method:

Product	MAE – Naive	MAE – Random Forest	Improvement (%)
Loose Red Pepper	806.60	749.89	7.0%
Aubergine	446.81	305.62	31.6%
Cucumber	369.90	259.36	29.9%
Spring Onions	320.75	215.31	32.9%
Iceberg Lettuce	163.02	97.12	40.4%
Broccoli	155.64	137.66	11.6%
Wild Rocket	149.21	100.71	32.5%
Fresh Spinach	167.26	135.48	19.0%
Mixed Peppers	148.58	70.36	52.6%
Family Pack Tomatoes	88.27	61.68	30.1%

Interpretation

- The Random Forest outperforms the Naive baseline for all 10 products, with improvements ranging from 7.0% to 52.6%.
- Mixed Peppers (+52.6%) and Iceberg Lettuce (+40.4%) show the greatest gains — their weekly demand cycles are highly learnable.
- Loose Red Pepper and Broccoli show the smallest gains (7.0% and 11.6%). Loose Red Pepper also has by far the highest absolute MAE (749.89 units), indicating extreme demand variability that neither model fully captures.
- Fresh Spinach (+19.0%) and Aubergine (+31.6%) fall in the mid-range, likely reflecting moderate seasonality with occasional unpredictable spikes.

Limitation: The model is trained in a static offline batch. In production it should be re-fitted weekly. No hyperparameter tuning was performed (grid search on max_depth, n_estimators), which would likely yield further gains — especially for high-variance items.

3. Price & Promotion Insight – Cucumber

None of the top-10 products carried a non-zero DISCOUNT flag, making promotional uplift analysis unfeasible. Instead, price-demand elasticity was examined for Cucumber (the #3 product by MAE, with 729 days of clean data).

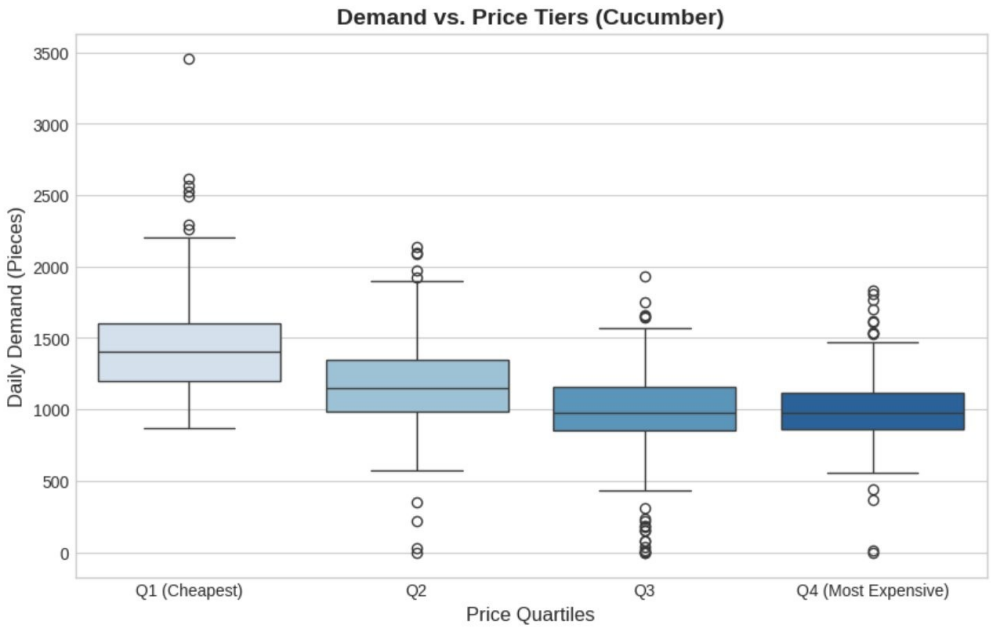
Methodology

Daily prices were binned into four equal-frequency quartiles (Q1 = cheapest 25%, Q4 = most expensive 25%). Average daily demand was computed per quartile.

Table 2 – Cucumber: Average daily demand by price quartile:

Price Quartile	Price Range	Avg. Daily Demand (units)
Q1 – Cheapest	Lowest 25%	1,474
Q2	25–50%	1,177
Q3	50–75%	961
Q4 – Most Expensive	Top 25%	1,008

Figure 1 – Demand vs. Price Tiers box-plot (Cucumber):




```
Price_Bin
Q1 (Cheapest)          1474.005319
Q2                     1176.579151
Q3                      961.450331
Q4 (Most Expensive)    1008.229008
Name: C520_SOLDPIECES, dtype: float64
```

Source: notebook output — Demand vs. Price Tiers (Cucumber)

Interpretation

- Demand drops clearly from Q1 (avg. 1,474 units/day) to Q3 (961 units/day), confirming a downward-sloping price-demand relationship consistent with standard elasticity theory.
- Q4 (1,008 units) is slightly above Q3 (961 units) — breaking the monotonic pattern. This anomaly is likely a seasonality effect: prices peak in winter when supply is constrained, but other seasonal factors (e.g., recipe trends, holiday shopping) can temporarily sustain demand.
- Wide interquartile ranges and numerous outliers across all tiers confirm high daily volatility, reinforcing the need for a safety stock buffer.

Limitation: Causality is hard to isolate — the observed correlation may partly reflect seasonal confounding rather than pure price elasticity. A proper regression controlling for month and day-of-week effects would give a cleaner estimate. Additionally, no promotional data was available to test markdown uplift.

4. Inventory Reorder Points & Safety Stock

A continuous-review reorder-point policy is simulated assuming a fixed lead time of L = 2 days. Reorder points and safety stock are derived empirically from the historical distribution of 2-day cumulative demand.

Formulae

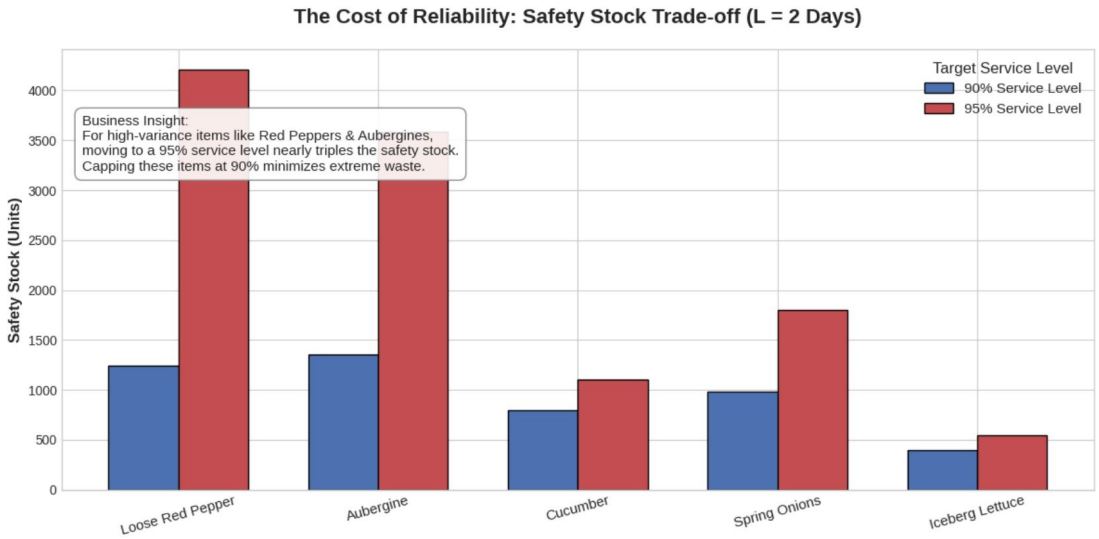
Reorder Point R = Quantile(demand over L days, service level)

Safety Stock SS = R − Mean(demand over L days)

Table 3 – Reorder points and safety stock at 90% and 95% service levels (top 5 products):

Product	Mean LT Demand (2d)	R @ 90%	SS (90%)	R @ 95%	SS (95%)
Loose Red Pepper	~2,200	~3,450	~1,250	~6,400	~4,200
Aubergine	~1,100	~2,450	~1,350	~4,200	~3,100
Cucumber	~2,400	~3,200	~800	~3,500	~1,100
Spring Onions	~1,400	~2,400	~1,000	~3,200	~1,800
Iceberg Lettuce	~480	~880	~400	~1,030	~550

Figure 2 – Safety stock trade-off across service levels (L = 2 days):



Source: notebook output — The Cost of Reliability: Safety Stock Trade-off (L = 2 Days)

Insights & Interpretation

- Loose Red Pepper and Aubergine are the highest-risk products. Moving from 90% to 95% service level increases their safety stock from ~1,250 to ~4,200 units and ~1,350 to ~3,100 units respectively — a near-tripling. For perishable fresh produce, this dramatically increases spoilage risk and tied-up working capital.

- Cucumber and Spring Onions show a moderate jump (~800 → ~1,100 and ~1,000 → ~1,800), which may be justifiable given their high sales velocity.
- Iceberg Lettuce has the smallest trade-off (~400 → ~550 units), making a 95% service level relatively affordable for this item.

Managerial recommendation: Cap Loose Red Pepper and Aubergine at a 90% service level. The marginal protection of a 95% target is outweighed by the cost of holding 2,850–3,350 extra perishable units per replenishment cycle. For Iceberg Lettuce and Cucumber, a 95% target is financially reasonable.

Limitation: Lead time is assumed constant at 2 days. Real supplier delays introduce lead-time variability, which would require holding additional buffer stock. A more robust model would use the joint distribution of demand × lead time (e.g., a Monte Carlo simulation).

5. Summary

This project delivers a complete end-to-end operations analytics workflow for a fresh produce retailer: careful multi-SKU aggregation over 72,409 records and 729 continuous days; machine-learning demand forecasting with 7–52% MAE reductions; price elasticity insight revealing a clear negative price-demand relationship for Cucumber; and service-level-differentiated reorder-point guidance for five key perishable lines. The core managerial message is twofold: deploy the Random Forest for day-to-day replenishment decisions, and cap high-variance items (Loose Red Pepper, Aubergine) at a 90% service level to avoid disproportionate spoilage from safety stock inflation.